

Réunion Louis + Nahuel + Konstantin



Seminar of the NEO team (Inria Sophia Antipolis)

Louis HAUSEUX (Inria Sophia Antipolis, superv. by K. AVRACHENKOV & J. ZERUBIA)

Joint work with Nahuel SOPRANO-LOTO (Inria Paris)

and Konstantin AVRACHENKOV (Inria Sophia Antipolis)

Inria Sophia Antipolis, Lagrange Gris room (L101) – 2.30 p.m. Thursday 16 July 2026



Outline

1. Motivation & the Bayesian framework
2. The Sankararaman & Baccelli model
3. MCMC Dynamics & Performance Bounds
4. Open Questions & Conclusion

01

Motivation & the Bayesian framework



Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

Haplotype 1
(Truth $\Sigma = +$)

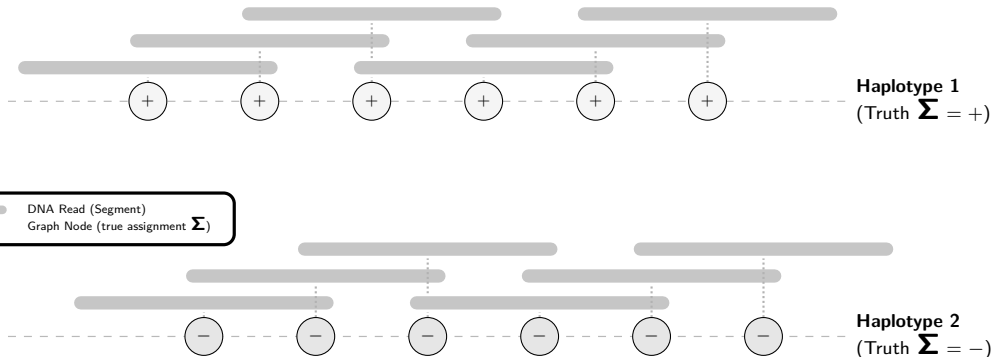
Haplotype 2
(Truth $\Sigma = -$)



Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.



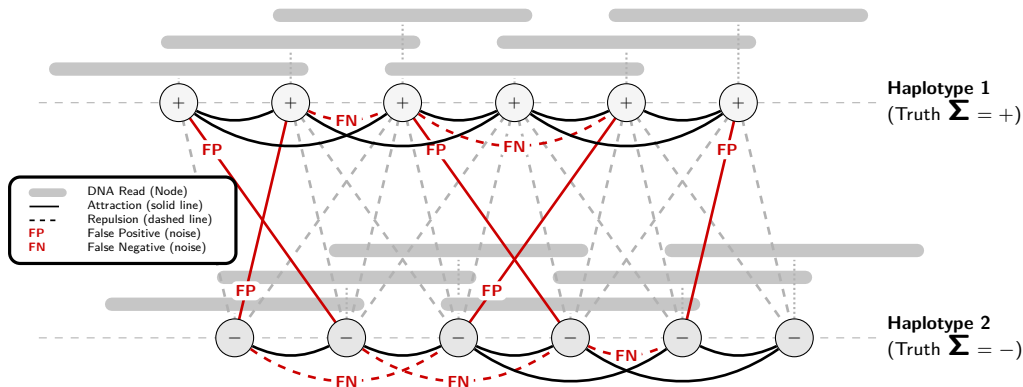


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads Edges = overlaps

Goal: Reconstruct the true haplotype assignments Σ from this observed signed graph.





Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- ▶ Partition nodes into $K = 2$ communities: $\Sigma_i \in \{+1, -1\}$.
- ▶ Links represent **attractions** (solid) and **repulsions** (dashed).
- ▶ Beyond the classical SBM: a rich framework coupling spatial geometry and community structure!

The Geometric Challenge

- ▶ Spatial embedding (e.g., positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- ▶ Edge probabilities depend on distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$

$$\mathbb{P}(\text{attractive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

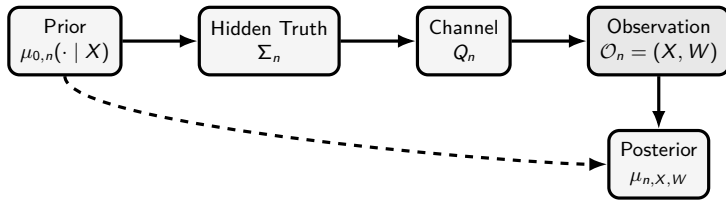
- ▶ **Main interest of the model... but also core challenge:** capturing this geometry.



The General Bayesian Model

Definition

1. **Nodes, Edges, Communities:** Nodes $V_n = \{1, \dots, n\}$, edges E_n , communities $\mathcal{L}_K = \{0, \dots, K-1\}$, and set of configurations $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$.
 2. **Features:** $X_n \in \mathcal{X}_n$ according to distribution ν_n .
 3. **Prior on configurations:** $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n$.
 4. **(Noisy) channel:** $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.
- The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x) \mu_{0,n}(\mathrm{d}\sigma \mid x) Q_n(\mathrm{d}w \mid x, \sigma)$.



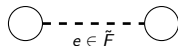
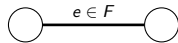


Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

Partition potential edges $E = F \dot{\cup} \tilde{F}$:

- ▶ F : **Attractive** (ferromagnetic), weight $W_e > 0$
- ▶ $\tilde{F}$: **Repulsive** (antiferromagnetic), weight $W_e < 0$



Signed Gibbs Measure [1]

Configuration probability $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

Gibbs Assumption

Special case of the Bayesian framework where the posterior is assumed to be:

$$\mu_{X,W}(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$$



Advantages of the Bayesian Framework

Rigorous Definitions

- ▶ **Algorithm** τ_n : MARKOV kernel mapping observations (X, W) to configurations.
- ▶ **Random Guess**: An algorithm agnostic to observations (relying only on the prior μ_0).
- ▶ **Weak / Partial Recovery**: Beating the best random guess asymptotically.

Posterior Resampling

Replace hidden truth by replica:

$$\Sigma \iff \Sigma^{(1)} \sim \mu(\cdot \mid X, W)$$

Measure of Performance: Overlap

The overlap quotients the HAMMING similarity by label permutations $\mathfrak{S}_K$:

$$\text{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}$$

- ▶ Uniform prior: best random guess score is $1/K$.
- ▶ Weak recovery: $\text{overlap} \geq 1/K + \eta$.



Posterior Resampling (Nishimori Identity)

Core Idea: Replacing Truth by a Replica

Comparing τ to truth $\Sigma \iff$ Comparing τ to replica $\Sigma^{(1)} \sim \mu$

Theorem (Nishimori Identity [2])

For any algorithm τ , posterior replica $\Sigma^{(1)} \sim \mu$, and bounded test function Ψ :

$$\mathbb{E} [\Psi(X, W, \Sigma, \tau)] = \mathbb{E} [\Psi(X, W, \Sigma^{(1)}, \tau)]$$

Corollary (Overlap Identity)

$$\mathbb{P} [\text{ov}(\Sigma, \tau) \geq s] = \mathbb{P} [\text{ov}(\Sigma^{(1)}, \tau) \geq s] \quad (\forall s \in [0, 1])$$

02

The Sankararaman & Baccelli model





The Sankararaman–Bacelli Model (GSBM)

Model Definition [3]

- **Nodes:** Realization X of a homogeneous POISSON point process in an n -volume of $\mathbb{R}^d$ [4].
- **Latent Communities:** Uniform assignments $\Sigma_i \in \{+1, -1\}$ ($K = 2$).
- **Edges:** Connection $e = \{i, j\} \in H$ occurs with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\text{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction),} \\ f_{\text{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion),} \end{cases}$$

where $r_e = \|X_i - X_j\|$ and $f_{\text{in}}(r) \geq f_{\text{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ_n (independent of Σ_n) achieving:

$$\lim_{n \rightarrow \infty} \mathbb{E}[\text{ov}_n(\Sigma_n, \tau_n)] \geq \frac{1}{2} + \eta \quad (\eta > 0), \quad \text{where } \text{ov}_n(\Sigma_n, \tau_n) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\tau_n, i = \pi(\Sigma_n, i)\}}.$$



GSBM in the Bayesian Framework

Sankararaman & Baccelli (GSBM)	Bayesian Framework
Positions X : Realization of a homogeneous POISSON point process Prior : Uniform prior μ_0 on $\Sigma \in \{+1, -1\}^n$ Observables : (H, X) where H is the observed graph	Features $X \sim \nu$: same POISSON law Prior μ_0 : same Observables : (W, X) where W is the channel output
Deterministic construction of W from H: $W_e = \begin{cases} \log \left(\frac{f_{\text{in}}(r_e)}{f_{\text{out}}(r_e)} \right) \geq 0 & \text{if } e \in H \text{ (attractive),} \\ -\log \left(\frac{1 - f_{\text{out}}(r_e)}{1 - f_{\text{in}}(r_e)} \right) \leq 0 & \text{if } e \notin H \text{ (repulsive).} \end{cases}$	
Posterior : $\mathbb{P}(\Sigma = \sigma \mid H, X)$	Posterior : $= \mathbb{P}(\Sigma = \sigma \mid W, X) = \mu(\sigma)$  GIBBS assumption



The Percolation Obstruction

Necessary Condition for Weak Recovery [3]

- Requires a global information flow graph to **percolate** [5] (possess a giant component).

Theorem (Sankararaman & Baccelli [3])

Independent bond percolation with edge probability

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

*yielding a giant component is a **necessary condition** for weak recovery.*

- **Our goal:** Retrieve and extend this result using unified **Bayesian MCMC coupling**.

03

MCMC Dynamics & Performance Bounds

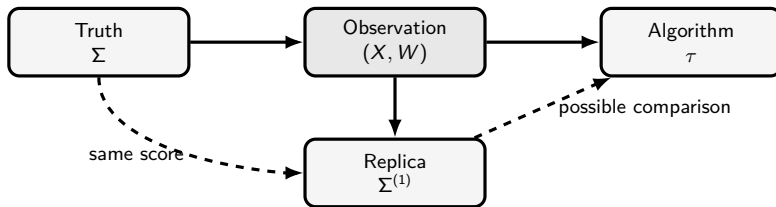




Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

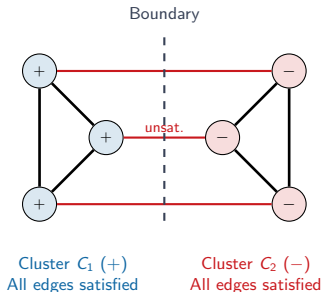
- ▶ We construct the posterior replica $\Sigma^{(1)}$ starting from the hidden truth Σ .
- ▶ Running **exactly one step** of a posterior-invariant MARKOV kernel M initialized at Σ yields a valid replica $\Sigma^{(1)} \sim M(\Sigma, \cdot)$.



Single-Spin Dynamics & Geometric Bottleneck

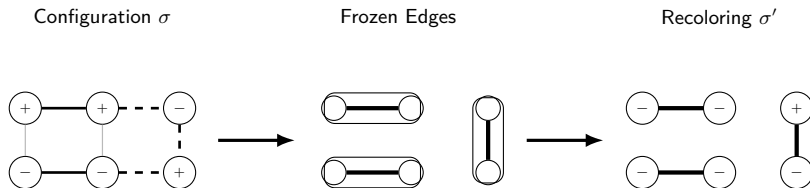
Single-Spin Update Limitations

- **Geometry makes updates slow:** Single-spin dynamics (GLAUBER / METROPOLIS-HASTINGS [6]) update one node at a time.
- **Contradictory Clusters:** Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- **High Energy Barrier:** Flipping a single boundary node violates internal edges, creating a local energy barrier.
- **Conclusion:** Need for geometry $\implies$ **Cluster Dynamics** (updating whole regions).





One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.



Swendsen–Wang Signed Cluster Dynamics

Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- ▶ Update whole clusters of vertices simultaneously to speed up mixing.
- ▶ **Freezing Step:** For each edge $e = \{i, j\}$:
 - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:
$$p_e^{\text{frozen}} = 1 - e^{-|W_e|}.$$
 - If e is unsatisfied, do not freeze it ($p_e^{\text{frozen}} = 0$).
- ▶ **Recoloring Step:** Flip each frozen cluster independently with probability $1/2$.



High-Order Triangular Dynamics

The Frustration Obstruction

- ▶ **Frustration:** $\nexists \sigma$ satisfying all edges (conflicting constraints).
- ▶ **Excessive Freezing:** In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- ▶ **Higher-Order Correlation:** Group interactions into **triangles** to correlate freezing and **freeze** as little as possible.

Triangular Cluster Dynamics [9]

- ▶ **Energy Decomposition:** Distribute edge energy U over triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- ▶ **Freezing Mechanism:** Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.



Triangular Dynamics Freezing Rules

$\triangle$ Attractive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	b_u	0	0	0	a_u
Other	1	0	0	0	0

$\triangleleft \triangle$ Repulsive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	1	0	0	0	0
$(+, +, -)$	b_u	0	$a_u/2$	$a_u/2$	0
$(+, -, +)$	b_u	$a_u/2$	0	$a_u/2$	0
$(+, -, -)$	b_u	$a_u/2$	$a_u/2$	0	0

Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$), where u is the energy transferred by each edge.

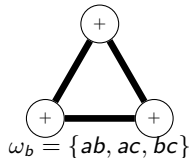
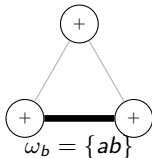
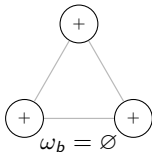


General Swendsen–Wang Cluster Dynamics: Framework

General Framework [10]

Suppose the Hamiltonian decomposes as: $U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma)$, where each $b \in \mathcal{B}$ is a **bond** (e.g., edge, triangle, clique).

1. **Freezing Step:** For each bond $b \in \mathcal{B}$, freeze a sub-bond $\omega_b \subseteq b$ with probability $P_b^\sigma(\omega_b)$.
2. **Recoloring Step:** Recolor each frozen cluster independently according to a distribution preserving the prior $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$.





General Swendsen–Wang Cluster Dynamics: Invariance

Definition (Local Reversibility)

For two configurations σ, σ' and a bond b , we define the compatibility probability:

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = \begin{cases} P_b^\sigma(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all $b, \omega_b, \sigma, \sigma'$:

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = e^{-(U_b(\sigma') - U_b(\sigma))} P_b^{\sigma' \rightarrow \sigma}(\omega_b)$$

Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and $P_b^\sigma(\emptyset) > 0$ for all b, σ :

1. The global cluster dynamics is reversible with respect to $\mu(\sigma) \propto e^{-U(\sigma)}$.
2. The condition $P_b^\sigma(\emptyset) > 0$ ensures the **irreducibility** of the MARKOV chain.

Percolation as a Necessary Condition for Recovery

Theorem (Percolation Bound on Overlap [10])

Under the uniform prior μ_0 , if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica $\Sigma_n^{(1)}$, any algorithm τ_n , and any $\eta > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta) \right] = 0$$

where $\theta^{\max}$ is the asymptotic fraction of nodes in giant frozen components (which bounds the maximum recoverable fraction of communities).

Corollary (Percolation is Necessary)

*If there is no percolation ($\theta^{\max} = 0$), the maximum overlap is bounded by $1/K$ (random guess): weak recovery is impossible. Hence, **percolation is a necessary condition for weak recovery**.*

Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.

Coupling & the Sankararaman–Baccelli Bound

One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth Σ . For any edge e , the probability that it is frozen is:

► **Case 1** ($\Sigma_i = \Sigma_j$):

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e) \left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e) \left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

► **Case 2** ($\Sigma_i \neq \Sigma_j$):

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e)) \left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e)) \left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

Corollary (Retrieving the S&B Bound [3])

*Under the classical edge SW dynamics, the frozen graph is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$



Strict Improvement of the S&B bound (on the triangular grid)

Homogeneous Model ($f_{\text{in}} = p$, $f_{\text{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW):** Frozen graph is independent edge percolation with parameter $p_{\text{edge}} = 2p - 1$. Critical threshold:

$$p_c^{\text{edge}} = \frac{1}{2} + \sin\left(\frac{\pi}{18}\right) \approx 0.673 \dots$$

- **Triangular Dynamics:** Frozen graph yields correlated triangle percolation. Critical threshold:

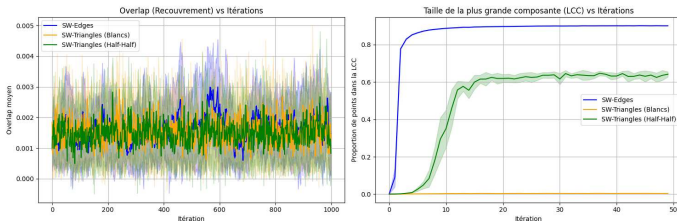
$$p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192 \dots$$

Corollary (Strictly Better Impossibility Bound [10])

Since $p_c^{\text{edge}} < p_c^{\triangle}$, for any $p \in [0.673 \dots, 0.7192 \dots)$, the standard S&B bound cannot conclude (edge dynamics percolate), whereas triangular dynamics do not, proving that weak recovery is impossible.



Empirical Results: Subcritical vs. Supercritical ($p = 0.70$)



Phase Space

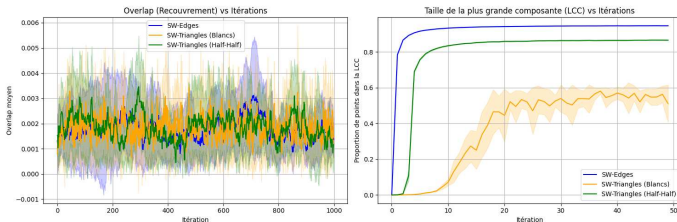
$(p_c^{\text{edge}} \approx 0.673 < 0.70 < p_c^{\Delta} \approx 0.7192)$

- ▶ **Edges (standard SW):** supercritical ($\theta^{\max} > 0$), largest component size (LCC) reaches ≈ 0.9 (right plot).
- ▶ **Triangles / Split weight:** subcritical ($\theta^{\max} = 0$), LCC remains near 0.

Observed Behavior

- ▶ Only standard edge SW achieves giant components.
- ▶ Overlap remains at the baseline noise level (around 0.002) for triangles and split weights due to lack of percolation (left plot).

Empirical Results: Onset of Percolation ($p = 0.72$)



Transition Regime

($p = 0.72 > p_c^\Delta \approx 0.7192$)

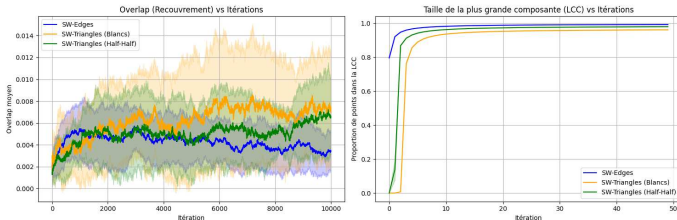
- ▶ All three dynamics are now **supercritical**.
- ▶ LCC grows for all: edges reach ≈ 0.9 , split weights reach ≈ 0.85 , and white triangles reach ≈ 0.55 (right plot).

Observed Behavior

- ▶ Giant components have formed, but the overlap remains low (below 0.003) for all dynamics (left plot).
- ▶ **No weak recovery yet:** the true weak recovery threshold is likely strictly above the percolation threshold (percolation is only a necessary condition).



Empirical Results: Weak Recovery Achieved ($p = 0.80$)



Supercritical Regime ($p = 0.80 \gg p_c^\Delta$)

- ▶ Giant components are well-established (right plot), with LCC close to 0.95 for edges and split weights, and 0.90 for white triangles.
- ▶ Fluctuations are small and components are stable.

Observed Behavior

- ▶ Weak recovery is successfully achieved for all algorithms.
- ▶ The overlap climbs significantly above the noise level (left plot), peaking at ≈ 0.010 for white triangles and ≈ 0.006 for split weights.



Exponential Clocks Replace Rigid Freezing

An edge $e = \{i, j\}$ is **satisfied** when

$$\sigma_i = \sigma_j, \quad e \in F,$$

$$\sigma_i \neq \sigma_j, \quad e \in \tilde{F}.$$

For each satisfied edge, independently,

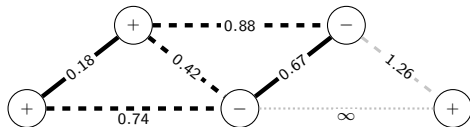
$$\xi_e \sim \text{Exp}(|W_e|).$$

At level t , keep the edges such that $\xi_e \leq t$; unsatisfied edges are never kept.

Classical Swendsen–Wang

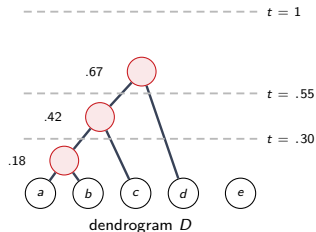
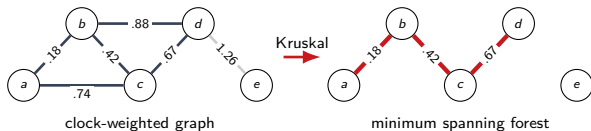
The usual freezing rule is exactly the cut $t = 1$:

$$\mathbb{P}(\xi_e \leq 1) = 1 - e^{-|W_e|}.$$





One Minimum Spanning Forest Contains Every Level



Exact pruning property

For every $0 \leq t \leq 1$, pruning the forest above level t gives exactly the connected components of the clock-weighted graph at level t :

$$\text{cc}\{e : \xi_e \leq t\} = \text{cc}\{e \in \text{MSF} : \xi_e \leq t\}.$$



Keep the Merge, Forget the Winning Edge

What the dendrogram D keeps

- ▶ the nested partitions of the vertices;
- ▶ the order and levels of the merges;
- ▶ the cut $t = 1$ separating the final forest components.

What D does not keep

The identity of the edge selected by Kruskal at a merge. Keeping it would hard-constrain the two children and reduce the update to ordinary component recolouring.

At a merge $C = C_1 \dot{\cup} C_2$, the exponential clocks form a race. After summing over the winning edge, its rate is simply

$$\text{rate}(C_1, C_2 \mid \sigma) = \sum_{\substack{e=\{i,j\}, i \in C_1, j \in C_2 \\ \sigma \text{ satisfies } e}} |W_e|.$$

Multiplying these exponential-race densities along D defines

$$\mathbb{P}(\mathrm{d}D \mid \sigma)$$

while marginalising the irrelevant winning edge. The same D may therefore have positive probability for σ and after a subtree flip. If not, the conditional heat bath assigns that flip probability zero.



An Edwards–Sokal Law on (σ, D)

Let $\mathbb{P}(\mathrm{d}D \mid \sigma)$ be exactly the clock–Kruskal law from the previous slides. On pairs (σ, D) , define

$$\mu(\mathrm{d}\sigma, \mathrm{d}D) = \frac{1}{Z} \mu_0(\mathrm{d}\sigma) e^{-U(\sigma)} \mathbb{P}(\mathrm{d}D \mid \sigma) = \mu(\mathrm{d}\sigma) \mathbb{P}(\mathrm{d}D \mid \sigma).$$

Correct marginal

Since $\mathbb{P}(\cdot \mid \sigma)$ is a probability law,

$$\int \mu(\mathrm{d}\sigma, \mathrm{d}D) = \frac{1}{Z} \mu_0(\mathrm{d}\sigma) e^{-U(\sigma)} = \mu(\mathrm{d}\sigma).$$

Conditional law used for recolouring

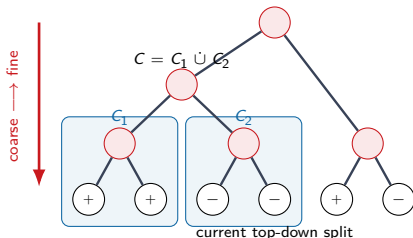
For a fixed dendrogram,

$$\mu(\mathrm{d}\sigma \mid D) \propto \mu_0(\mathrm{d}\sigma) e^{-U(\sigma)} \mathbb{P}(\mathrm{d}D \mid \sigma).$$

The prior μ_0 is therefore preserved exactly; it need not be uniform.



Prune from Coarse to Fine, with Two Choices at Each Split



At the current split $C = C_1 \dot{\cup} C_2$, compare only two configurations:

- the current configuration σ ;
- σ^{C_1} , obtained by reversing every label in C_1 .

Replace σ by σ^{C_1} with the conditional heat-bath probability

$$\frac{\mu(\sigma^{C_1} | D)}{\mu(\sigma | D) + \mu(\sigma^{C_1} | D)}.$$

Then update σ , descend to the next split, and recompute the probability.

The essential point

The choices at different levels are sequential conditional updates, not independent flips.

The Gibbs Measure Is Stationary by Construction

One split satisfies detailed balance

For the two configurations σ and σ^C ,

$$\mu(\sigma \mid D) \mathbb{P}(\sigma \rightarrow \sigma^C \mid D) = \frac{\mu(\sigma \mid D) \mu(\sigma^C \mid D)}{\mu(\sigma \mid D) + \mu(\sigma^C \mid D)} = \mu(\sigma^C \mid D) \mathbb{P}(\sigma^C \rightarrow \sigma \mid D).$$

Hence every split preserves $\mu(\cdot \mid D)$, and so does the complete top-down sweep.

$$\sigma \xrightarrow{D \sim \mathbb{P}(\cdot \mid \sigma)} (\sigma, D) \xrightarrow{\text{top-down heat baths}} (\sigma', D) \xrightarrow{\text{discard } D} \sigma'.$$

Whole transition

The first arrow samples one conditional law of $\mu(d\sigma, dD)$; the sweep preserves the other conditional law. This is a Gibbs-within-Gibbs update on (σ, D) . Therefore

$$\sigma \sim \mu \implies \sigma' \sim \mu.$$

04

Open Questions & Conclusion





Open Questions

Limits of the One-Replica Coupling

- ▶ Our framework constructs **one** replica $\Sigma_n^{(1)}$ using MCMC cluster dynamics.
- ▶ This only yields an impossibility bound (**necessary condition** for recovery).

Towards a Necessary and Sufficient Condition

- ▶ Achieving a **necessary and sufficient** condition requires a **two-replica** framework.
- ▶ Open: How to define and analyze two-replica couplings under MCMC?

Towards a Practical Community Detection Algorithm

Limits of Direct Gibbs Sampling

- ▶ Direct posterior sampling does not perform well in practice.
- ▶ Likely due to slow mixing of the underlying MARKOV chain.

Estimating the Correlation Matrix

- ▶ Estimating the correlation matrix $\mathbb{P}(\sigma_i = \sigma_j)$ yields much better results.
- ▶ **Future work:** Understand why correlation estimators are more robust.



Distance Adapted to Cluster Dynamics

Hamming Similarity on the Quotient Space

- ▶ Standard overlap is a **HAMMING** similarity quotiented by label permutations:

$$\text{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}.$$

Markov Transition-Adapted Distance

- ▶ Consider distances reflecting transitions of the cluster dynamics.
- ▶ For standard SW, this corresponds to the **symmetric difference** on satisfied edges:

$$d_{\text{sym}}(\sigma, \sigma') = \frac{1}{2} \sum_{e=\{i,j\}} |W_e| \left| \mathbb{I}_{\{\sigma \text{ satisfies } e\}} - \mathbb{I}_{\{\sigma' \text{ satisfies } e\}} \right|$$

- ▶ Establishes links to **spectral graph analysis** (e.g., LAPLACIAN eigenvalues).

Application: 3-SAT Solving via Signed Graphs

The 3-SAT Problem

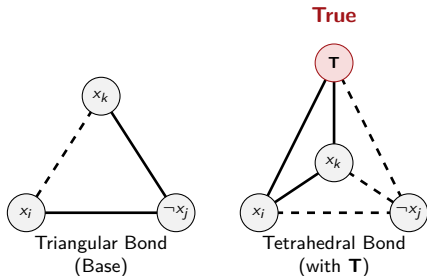
Find a truth assignment satisfying a set of clauses, e.g.,

$$\mathcal{C} = x_i \vee \neg x_j \vee x_k$$

Clause Encoding

We introduce a reference node **T** (True):

- ▶ **Triangle** (literals $x_i, \neg x_j, x_k$): edge signs opposite to the literal product.
- ▶ **Tetrahedron (with T)**: edge signs conforming to the literal signs.



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Merci.

